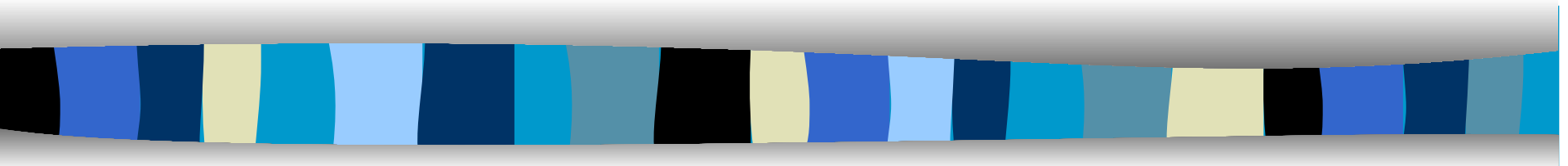


Asymmetric Cryptography



RSA, Elgamal

Problems with Symmetric Cryptography

- Secret key must be shared among communicants through secure medium such as the use of a key distribution center.
- The sender cannot sign the document digitally with symmetric cryptography.
- To solve these problems, we have asymmetric cryptography.

Misconception on Symmetric Crypto

- First misconception is that public-key encryption is more secure from cryptanalysis than is symmetric encryption.
- In fact, the security of any encryption scheme depends on the length of the key.
- A second misconception is that public-key encryption has made symmetric encryption obsolete.
- However, asymmetric has more computational overhead.

Misconception on Symmetric Crypto

- Some applications of symmetric cryptography needs central agent to manage keys which is not simpler and efficient than symmetric cryptography.

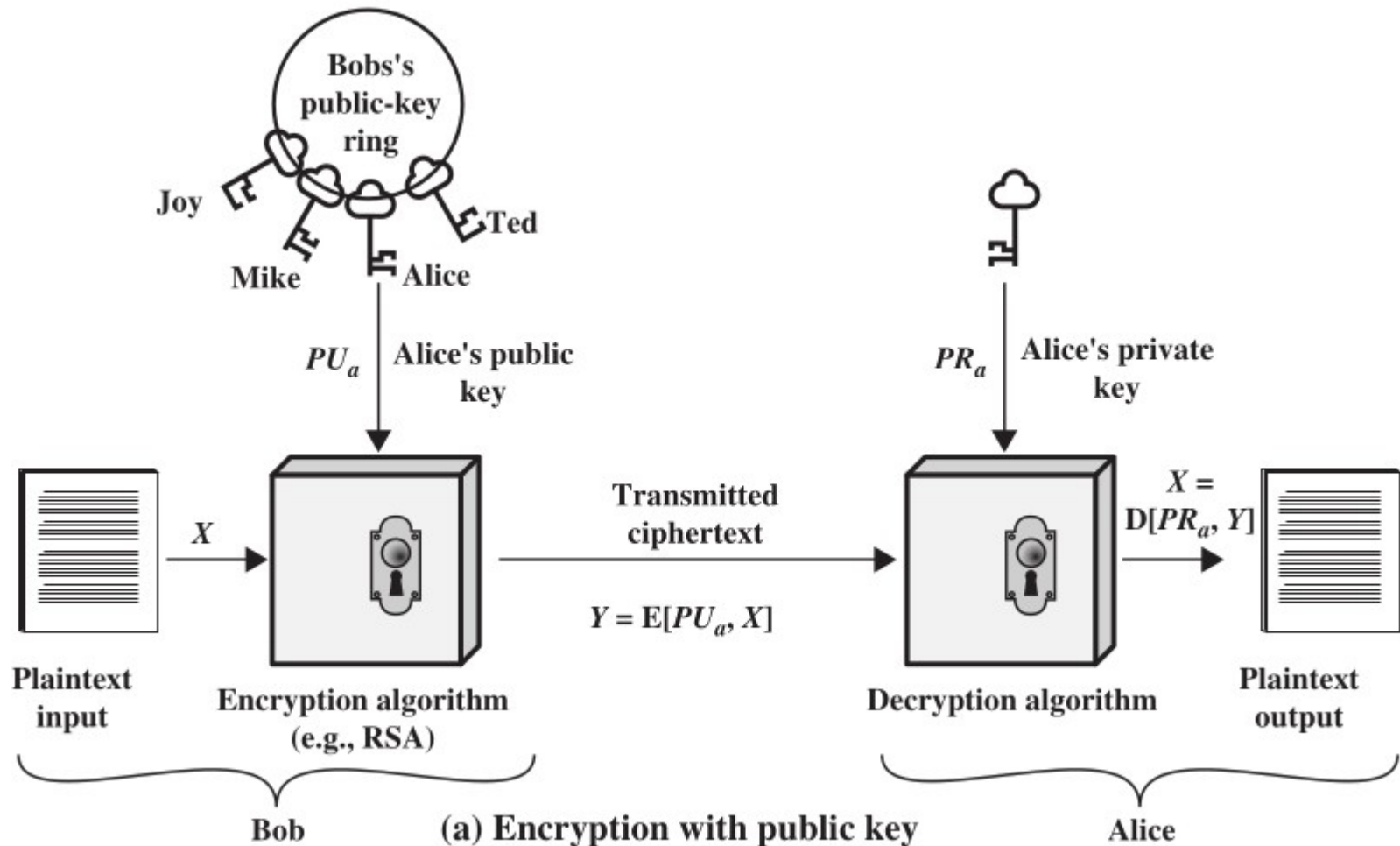
Asymmetric Cryptography

- In Asymmetric cryptography, each user generates a pair of keys to be used for the encryption and decryption of messages.
- Each user shares public key through in a public register or other accessible file.
- The companion key is kept private.
- Each user maintains a collection of public keys obtained from others.
- If Bob wishes to send a confidential message to Alice, Bob encrypts the message using Alice's public key.

Asymmetric Cryptography

- When Alice receives the message, she decrypts it using her private key.
- No other recipient can decrypt the message because only Alice knows Alice's private key.
- So, asymmetric cryptography uses different keys for encryption and decryption.
- The use of two keys has profound consequences in the areas of confidentiality, key distribution, and authentication

Asymmetric Cryptography



Public Key Cryptography

- Asymmetric cryptography is also known as public key cryptography.
- It is because Asymmetric cryptography publicly shares the public key.
- The development of public-key cryptography is the greatest revolution in the entire history of cryptography.
- Public-key algorithms are based on mathematical functions rather than on substitution and permutation.

Characteristics of public crypto

- It is computationally infeasible to determine the decryption key given only knowledge of the cryptographic algorithm and the encryption key.
- Either of the two related keys can be used for encryption, with the other used for decryption.

Ingredients of public cryptography

- **Plaintext:** This is the readable message or data that is fed into the algorithm as input.
- **Encryption algorithm:** The encryption algorithm performs various transformations on the plaintext.
- **Public and private keys:** This is a pair of keys that have been selected so that if one is used for encryption, the other is used for decryption.
- **Ciphertext:** This is the scrambled message produced as output.
- **Decryption algorithm:** produces the original plaintext using the matched key.

Applications of public cryptography

- **Encryption/decryption:** The sender encrypts a message with the recipient's public key.
- **Digital signature:** The sender “signs” a message with its private key. Signing is achieved by a cryptographic algorithm applied to the message or to a small block of data that is a function of the message.
- **Key exchange:** Two sides cooperate to exchange a session key. Several different approaches are possible, involving the private key(s) of one or both parties.

Applications of public cryptography

- Some public cryptography with their applications
- Digital Signature Standard

Algorithm	Encryption/Decryption	Digital Signature	Key Exchange
RSA	Yes	Yes	Yes
Elliptic Curve	Yes	Yes	Yes
Diffie-Hellman	No	No	Yes
DSS	No	Yes	No

Requirements for Public Crypto

- It is computationally easy for to generate a pair (public key Pub, private key PRb).
- It is computationally easy for a sender A, knowing the public key and the message to be encrypted, M, to generate the corresponding ciphertext:
- $C = E(\text{PUb}, M)$
- It is computationally easy for the receiver B to decrypt the ciphertext using the private key
- $M = D(\text{PRb}, C) = D[\text{PRb}, E(\text{PUb}, M)]$

Requirements for Public Crypto

- It is computationally infeasible for an adversary, knowing the public key, Pub , to determine the private key, PRb .

Cryptanalysis of Public Key Crypto

- Use of large key in public cryptography to make brute force attack infeasible makes it very slow.
- So, public cryptography is confined to key management and signature applications.
- Computing private key given public key is infeasible.
- If DES secret key is being shared using public cryptography then, attacker can encrypt all possible 56 bit keys using public key.
- If encryption matched with the ciphertext then attacker will know the DES key.

RSA

- The most common public-key algorithm is the RSA cryptosystem, named for its inventors (Rivest, Shamir, and Adleman).
- RSA uses two exponents, e and d , where e is public and d is private.
- Suppose P is the plaintext and C is the ciphertext. Alice uses $C = P^e \bmod n$ to create ciphertext C from plaintext P . Here e is public key of Bob.
- Bob uses $P = C^d \bmod n$ to retrieve the plaintext sent by Alice. Where d is private key of Bob.

RSA

- The modulus n , a very large number, is created during the key generation process.
- Encryption and decryption use modular exponentiation.
- Exponentiation takes $O((\log n)^3)$ operations (easy)
- For decryption, attacker must know private key d .
- Which is one of the factor of n .
- Factorization takes $O(e \log n \log \log n)$ operations (hard)

RSA: Key generation

- Select large distinct primes p and q using primality test.
- $n=pq$
- $\phi(n)=(p-1)(q-1)$
- Select positive integer e such that $\gcd(e, \phi(n))=1$
- Select d such that $ed \equiv 1 \pmod{\phi(n)}$
- Public key $\{e, n\}$
- Private key $\{d, n\}$

RSA: Encryption

- m is the plaintext
- Value of m is between 0 to $n-1$
- $C = m^e \bmod n$

RSA: Decryption

- C is the plaintext
- Value of C is between 0 to $n-1$
- $m = C^d \bmod n$

RSA: Correctness of Decryption

- $C^d \bmod n$
- $= (m^e)^d \bmod n$
- $= m^{ed} \bmod n$
- $= m^{q\phi(n)+1} \bmod n$
- $= m^{q\phi(n)} \times m \bmod n$
- $= [(m^{\phi(n)} \bmod n)^q (m \bmod n)] \bmod n$
- $= [1^q (m \bmod n)] \bmod n$
- $= [1 m] \bmod n = m$

RSA example: Key Generation

- Select $p=17$, $q=11$
- $n=pq=17 \times 11=187$
- $\phi(n)=(p-1)(q-1)=16 \times 10=160$
- Select $e=7$ as $\gcd(\phi(187), 7)=1$
- Compute inverse of e using extended euclidean algorithm
- $ed \equiv 1 \pmod{\phi(n)}$

RSA example: Key Generation

q	n	b	r	t1	t2	t=t1-t2*q
22	160	7	6	0	1	-22
1	7	6	1	1	-22	23
6	6	1	0	-22	23	-160
	1	0		23	-160	

- So, multiplicative inverse of e (7) is $d(23)$
- Private key is $\{23, 187\}$
- Public key is $\{7, 187\}$

RSA example: Encryption

- Let plaintext, $m=118$, $e=7$, $n=187$
- $C=m^e \bmod n = 118^7 \bmod 187$
- Here, 7 in binary is 111 which is 4,2,1
- $=118^4 \times 118^2 \times 118 \bmod 187$
- $118^2 \bmod 187 = 13924 \bmod 187 = 86$
- $118^4 \bmod 187 = (118^2)^2 \bmod 187 = (118^2 \bmod 187)^2 \bmod 187$
- $= 86^2 \bmod 187 = 7396 \bmod 187 = 103$
- Now, $118^4 \times 118^2 \times 118 \bmod 187 = 103 \times 86 \times 118 \bmod 187$
- $= 8858 \times 118 \bmod 187 = 69 \times 118 \bmod 187 = 8142 \bmod 187 = 101$

RSA example: Decryption

- Ciphertext is $C=101$, $d=23$, $n=187$
- $m=C^d \bmod n = 101^{23} \bmod 187$
- 23 in binary is 10111 which is 16,4,2,1
- $101^{23} \bmod 187 = 101^{16} \times 101^4 \times 101^2 \times 101 \bmod 187$
- $= 101^2 \bmod 187 = 10201 \bmod 187 = 103$
- $= 101^4 \bmod 187 = (101^2)^2 \bmod 187 = (101^2 \bmod 187)^2 \bmod 187$
- $= 103^2 \bmod 187 = 10609 \bmod 187 = 137$
- $= 101^8 \bmod 187 = (101^4)^2 \bmod 187 = (101^4 \bmod 187)^2 \bmod 187$
- $= 137^2 \bmod 187 = 18769 \bmod 187 = 69$

RSA example: Decryption

- $101^{16} \bmod 187 = (101^8)^2 \bmod 187 = (101^8 \bmod 187)^2 \bmod 187$
- $= 69^2 \bmod 187 = 4761 \bmod 187 = 86$
- Now, $101^{23} \bmod 187 = 101^{16} \times 101^4 \times 101^2 \times 101 \bmod 187$
- $= 86 \times 137 \times 103 \times 101 \bmod 187$
- $= [(86 \times 137 \bmod 187) \times (103 \times 101 \bmod 187)] \bmod 187$
- $= 1 \times 118 \bmod 187$
- $= 118$

Elgamal Cryptosystem

- Elgamal is another public-key cryptosystem.
- It is named after the inventor, Taher ElGamal.
- ElGamal is based on the discrete logarithm problem.

Elgamal Encryption

- 1. Select large prime p and its primitive root e_1 in group $G = \langle \mathbb{Z}_{p^*}, \times \rangle$
- 2. Generate random integer d such that $1 \leq d \leq p-2$
- 3. Compute $e_2 = e_1^d \bmod p$
- 4. Select random integer r such that $1 < r < p-1$
- 4. Now compute initial cipher text c_1
 $c_1 = e_1^r \bmod p$
- 5. Now compute actual cipher text c_2
 $c_2 = m \times e_2^r \bmod p$

Elgamal: Decryption

- $m = [c_2 (c_1^d)^{-1}] \bmod p$
- Correctness
- $[C_2 \times (C_1^d)^{-1}] \bmod p$
- $= [(e_2^r \times m) \times (e_1^{rd})^{-1}] \bmod p$
- $= (e_1^{dr}) \times m \times (e_1^{rd})^{-1} \text{ as } e_2 = e_1^d \bmod p$
- $= m$
- Elgamal uses the property that given e_2 , e_1 , and p , it is infeasible to calculate $r = \text{dlog}_{e_1} e_2 \bmod p$

Elgamal Example: Key Generation

- Lets use Z_{19} , $p=19$, $e_1=10$, $d=5$
- $e_2=e_1^d \bmod 19$
- $=10^5 \bmod 19$
- $=3$
- Private key $d=5$
- Public key $e_1=10$, $e_2=3$, $p=19$

Elgamal Example: Encryption

- Public key $e_1=10$, $e_2=3$, $p=19$, $m=17$
- Select random integer $r=6$ in Z_{19}
- $c_1=e_1^r \bmod p$
- $=10^6 \bmod 19=11$
- $c_2=m \times e_2^r \bmod p$
- $=17 \times 3^6 \bmod 19$
- $=17 \times 7 \bmod 19$
- $=5$

Elgamal Example: Decryption

- private key $d=5$, $c_1=11$, $c_2=5$, $p=19$
- $m=c_2 \times (c_1^d)^{-1} \bmod p$
- $=5 \times (11^5)^{-1} \bmod 19$
- $=[5 \times (11^5 \bmod 19)^{-1}] \bmod 19$
- 5 in binary is 101 which is 4, 1
- $[5 \times (11^4 \times 11 \bmod 19)^{-1}] \bmod 19$
- $11^2 \bmod 19=7$
- $11^4 \bmod 19=(11^2)^2 \bmod 19=7^2 \bmod 19=11$

Elgamal Example: Decryption

- Now, $[5 \times (11^4 \cdot 11 \bmod 19)^{-1}] \bmod 19$
- $= [5 \times (11 \times 11 \bmod 19)^{-1}] \bmod 19$
- $= [5 \times (7)^{-1}] \bmod 19$
- $= 5 \times 11 \bmod 19 = 17$

Distribution of Secret Key using Public cryptography

- Two communicant must share the secret key before they actually communicate using symmetric cryptography.
- We can use public key cryptography to share the secret key of symmetric cryptography.
- One such method is Diffie-Hellman protocol.

Diffie-Hellman Protocol

- Two communicant must share the secret key before they actually communicate using symmetric cryptography.
- We can use public key cryptography to share the secret key of symmetric cryptography.
- One such method is Diffie-Hellman protocol.
- In the Diffie-Hellman protocol two parties choose two numbers p and g .
- The first number p is a large prime number on the (1024 bits).
- The second number, g , is a generator of order $p - 1$ in the group $\langle \mathbb{Z}_p^*, \times \rangle$.

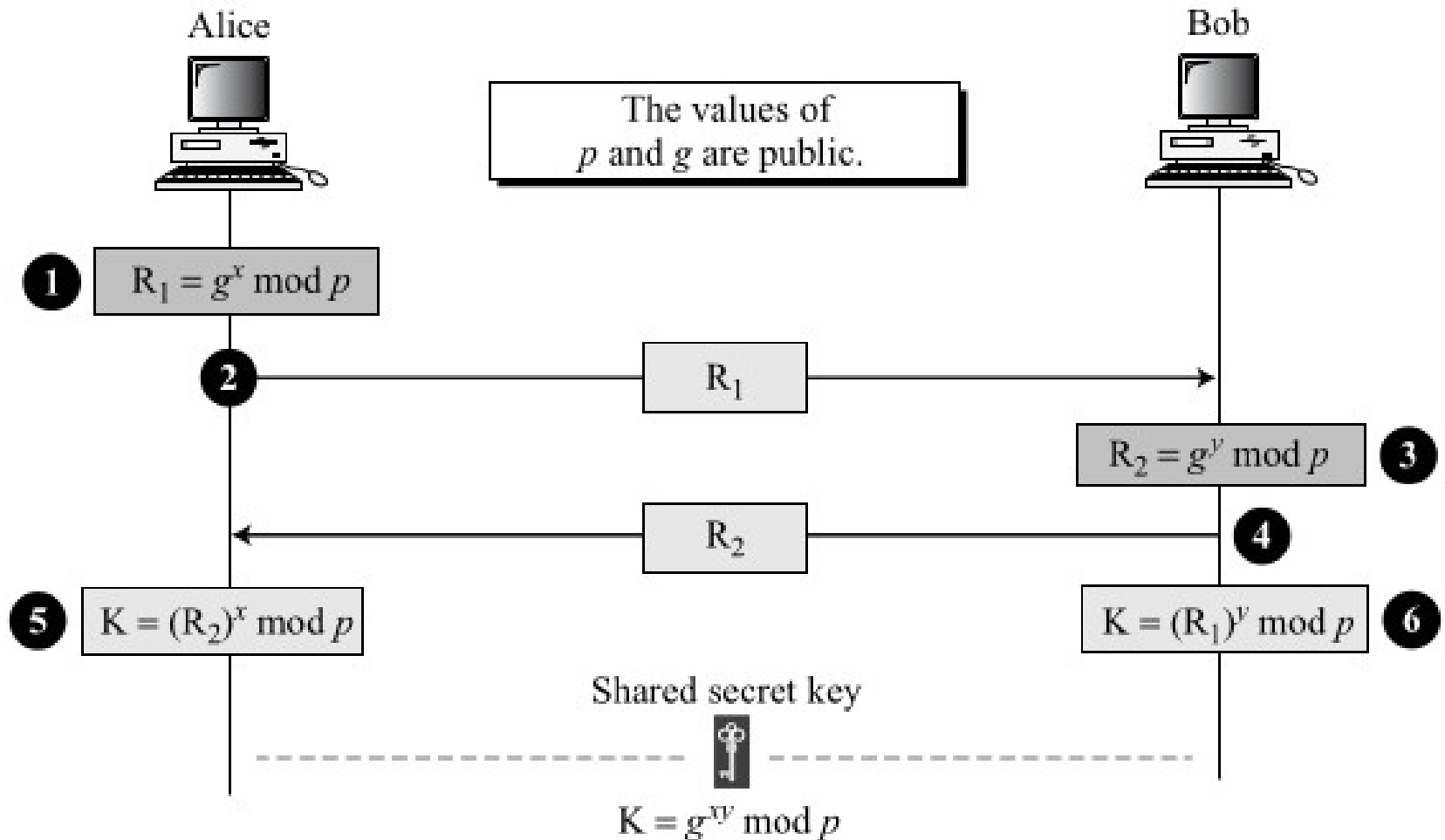
Diffie-Hellman Protocol

- Suppose Alice and Bob are communicating each other.
- 1. Alice and Bob agrees upon a large prime number p and a prime number g .
- 2. Alice chooses a large random number x such that $0 \leq x \leq p-1$ and calculate $R_1 = g^x \bmod p$.
- 3. Bob also chooses another large random number y such that $0 \leq y \leq p-1$ and calculate $R_2 = g^y \bmod p$.

Distribution of Secret Key using Public cryptography

- 5. Alice sends R_1 to Bob.
- 6. Bob sends R_2 to Alice.
- 7. Alice calculates shared key, $K = (R_2)^x \bmod p$.
- 8. Bob also calculates shared key, $K = (R_1)^y \bmod p$.
- Both K is same because
- $K = (R_2)^x \bmod p = (g^y \bmod p)^x \bmod p = g^{yx} \bmod p$

Distribution of Secret Key using Public cryptography



Diffie Hellman Example

- Alice and Bob, $p=19$, $g=5$
- Alice, $x=6$
- Alice, $R_1 = g^x \bmod p = 5^6 \bmod 19 = 7$
- Bob, $y=7$
- Bob, $R_2 = g^y \bmod p = 5^7 \bmod 19 = 16$
- Alice Key, $K = R_2^x \bmod p = 16^6 \bmod 19 = 7$
- Bob key, $K = R_1^y \bmod p = 7^7 \bmod 19 = 7$

Man in the middle attack

- 1. Alice chooses x , calculates $R1 = g^x \bmod p$, and sends $R1$ to Bob.
- 2. Eve, the intruder, intercepts $R1$. She chooses z , calculates $R2 = g^z \bmod p$, and sends $R2$ to both Alice and Bob.
- 3. Bob chooses y , calculates $R3 = g^y \bmod p$, and sends $R3$ to Alice.
- $R3$ is intercepted by Eve and never reaches Alice.
- 4. Alice and Eve calculate $K1 = g^{xz} \bmod p$, which becomes a shared key between Alice and Eve.

Man in the middle attack

- Alice, however, thinks that it is a key shared between Bob and herself.
- 5. Eve and Bob calculate $K2 = g^{zy} \bmod p$, which becomes a shared key between Eve and Bob.
- Bob, however, thinks that it is a key shared between Alice a

Man in the middle attack

